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Roll No

EC-7002-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Satellite Communication

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume suitable data, if required.

1. a) Write short note on frequency allocations for satellite systems.
b) Write and explain Kepler's three laws of planetary motion.
2. a) What is meant by look angles? Explain them with reference to a geostationary satellite and the earth station.
b) Explain briefly what is meant by sun transit outage.
3. a) With the help of a suitable block diagram explain the function of Telemetry, Tracking and Command (TTC) sub-system.
b) Write short note transmit-receive earth station.
4. a) Derive general link equation. Find out expression for C/N and G/T ratio?
b) The EIRP of a 240W transponder is 57 dB W. Calculate the approximate gain of the antenna if the transponder is switched to 120W. Also calculate the new EIRP assuming that the same antenna is used.

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5. a) With the help of suitable diagram explain the satellite communication system architecture using VSAT. Also write the applications of VSAT.
b) With the help of suitable diagram describe the working of home receiver units in DBS television.
6. a) Describe in brief different types of antennas used in satellite communication.
b) A satellite carrying a 11.7GHz continuous wave (CW) beacon transmitter is located in a geostationary orbit 38,000km from an earth station. The beacon's output power is 200mW and it feeds an antenna with 18.9 dB gain towards the earth station. The earth station receiving antenna aperture efficiency is 50 percent. Calculate the path loss in dB and receiving antenna gain in dB.
7. a) What are the various interferences that may affect the satellite link performance? Explain in short.
b) Three amplifiers are connected in cascade having a gain of 20 dB each. If the noise temperature is 100K, determine the overall gain.
8. a) Write short note on history of development of satellite launch vehicles.
b) Satellite-1 in an elliptical orbit has the orbit semi major axis equal to 18,000km and satellite-2 has semi major axis equal to 24,000km. Determine the relationship between their orbital periods.
